

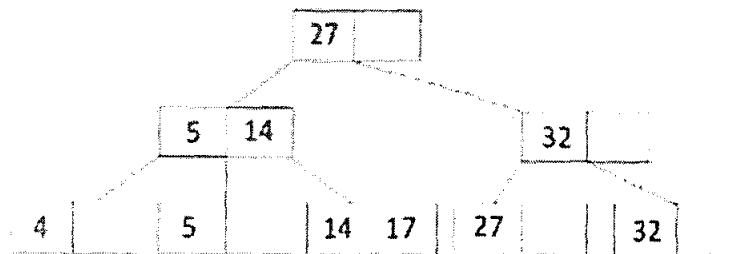


INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
End-Spring Semester Examination 2023-24

Date of Examination: _____ Session: (FN/AN) _____ Duration: **3 hours** Full Marks: **50**
 Subject No.: **MA60308** Subject: **FILE ORGANIZATION AND DATABASE SYSTEMS**
 Department/Center/School: **Mathematics**
 Specific charts, graph paper, log book, etc., required: **NO**
 Special Instructions (if any): Answer **ALL EIGHT (8)** Questions. This question paper consists of **3** pages.

1.

- I) Construct a B+-tree for the following set of key values:
 (1, 3, 5, 7, 9, 2, 4, 6, 8, 10)
 Assume that the tree is initially empty and values are added in the same order as given.
 Suppose each node of this B+ tree can hold up to 4 pointers and 3 keys.
- II) Remove the keys (9, 7, 8) in this order and show the tree after each step.
- III) What is the maximum number of key values we can add to the following B+ tree without increasing its height?



[3+2+2]

2.

- I) Explain the ACID Properties of transaction with an example.
- II) Consider the following schedule S for 6 transactions T_1, T_2, T_3, T_4, T_5 , and T_6 . Check whether the schedule is conflict serializable or not.

$R_2(A); W_3(A); R_4(B); W_4(B); W_5(A); W_5(B); W_1(A); R_1(C); W_6(C);$

[4+3]

3.

- I) Show that there are schedules that are possible under the two-phase locking protocol, but are not possible under the timestamp protocol, and vice versa.
- II) In the context of Deadlock Avoidance, compare the 'Wait-Die' and 'Wound-Wait' algorithms using an appropriate example.

[3+3]

(P.T.O.)

4.

I) Consider the following CLINIC database:

Patients (p_id, p_name, age, p_email)
 Doctors (d_id, d_name, qualification, d_email)
 Appointments (p_id, d_id, date, time)

Write the most optimized relational algebra for the following query using heuristic based query optimization technique.

SELECT Patients.p_name, Doctors.d_name
 FROM Patients, Doctors, Appointments
 WHERE Patients.p_id=Appointments.p_id AND Doctors.d_id=Appointments.d_id AND
 Patients.age=50;

II) Consider the relations $R_1(A, B, C)$, $R_2(C, D, E)$, and $R_3(E, F)$, with primary keys A, C, and E, respectively. Assume that R_1 has 1000 tuples, R_2 has 1500 tuples, and R_3 has 750 tuples. Estimate the size of $R_1 \bowtie R_2 \bowtie R_3$, and give an efficient strategy for computing the join.

[3+2]

5.

I) Suppose that we are using hashing on a file that contains the following records.

Acc. Number	Name	Balance
A-217	Rahul	7000.00
A-101	Raghav	4500.00
A-110	Priya	300.00
A-215	Amit	3000.00
A-102	Sunny	5700.00
A-201	Mehul	8600.00
A-218	Shakshi	2000.00
A-222	Rohit	150.00
A-305	Tanmay	1540.00

Show the hash structure for this file if the hash function is $h(x) = \text{Sum of the digits of the Acc. Number} \bmod 7$. Assume that the buckets are of size 2 each. Do you face any problem in creating the hash structure? If yes, mention it and suggest a solution.

II) Give an example of Extendible Hashing and a list of entries e_1, e_2, e_3 such that inserting the entries in order leads to three splits and deleting them in the reverse order yields the original index. If such an example does not exist, explain.

[3+3]

(P.T.O.)

6.

- I) When is a system considered 'Reliable'? Discuss the difference between the 'Forward-Error' and 'Backward-Error' recovery schemes of a Reliable system.
- II) Explain how the 'Write-Ahead Logging Protocol' is useful in recovery from a system failure.
- III) Explain different types of fragmentation in DDBS with an appropriate example. Why data replication is useful in DDBS?

[(1+2)+2+(3+1)]

7. Consider the following PROJECT database.

Project (Pj_ID, Project_name, Budget, Start_date, Status)
Professor (Prof_ID, Prof_name, Department, E-mail)
Student (Roll_no, Student_name, Department, E-mail, Hall)
Working (Roll_no, Pj_ID, Prof_ID, Stage, Status)

Express the following queries in SQL:

- I) Find the professors' email address who are guiding the project work which is in planning stage, working status is In-progress and students from MMM hall are involved in the project.
- II) Find the professors' ID, the total budget, and the average budget of the projects guided by the corresponding professor.
- III) Find the names and e-mail addresses of students and professors from 'Mathematics' department and order them by their names.

[3*2]

8. Consider the following dataset:

T ID	Items
T ₁	I1, I3, I4
T ₂	I2, I3, I5, I6
T ₃	I1, I2, I3, I5
T ₄	I2, I5
T ₅	I1, I3, I5

Use **Apriori** algorithm to find the candidate and frequent item sets. Enumerate all the final frequent item sets. Also, derive the association rules that are generated. Assume minimum support count, $S = 2$ and minimum confidence, $C = 75\%$.

[4]

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